

YASH GANDHI

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EXPERIENCE

Research Engineer, Collaboratory for Advanced Computing and Simulations

07/2023 - Present

- Developed **diffusion-based generative AI** solution to generate peptide sequences for cancer research, achieving 88% accuracy in predicting their binding effectiveness.
- Built end-to-end **multi-agent AI workflow** using LangGraph, LangChain, and PydanticAI to automate peptide generation, screen and rank candidates, and manage reproducible model experiments.
- Implemented **self-supervised DnCNN** model to denoise microscopy images, improving the signal-to-noise ratio by 36% and revealing hidden features in nanoparticle images.
- Integrated the model into **real-time inference AWS pipeline** using SageMaker, Lambda, and S3, with web interface that enabled researchers to upload, process, and analyze images while reducing workflow overhead by 40%.
- Deployed **Probabilistic Deep Learning, Transformer and Liquid neural network** forecasting methods to enhance accuracy of oil production estimates at petroleum wells from 30% to 78%.

Software Engineer, Intern, Alation

05/2022 - 08/2022

- Engineered Configuration Management System (CMS) to facilitate customer migration to **multi-tenant cloud architecture**.
- Utilized Django-gRPC with AWS RDS PostgreSQL to build scalable APIs for CRUD operations on configuration data.
- Designed Redis caching across the CMS client and server layers, reducing API latency by 40 - 60%.
- Integrated CMS with encryption microservice to secure customer configuration data at rest and implemented Datadog monitoring for API performance tracking.

Software Developer, Intern, Empays Payment Systems

03/2020 - 08/2020

- Developed **Conversational banking chatbot** using Google Dialogflow to automate retrieval of transaction history, card details, and account summaries with dynamic user filters.
- Implemented intent recognition, entity extraction, and iterative training on real user queries to enhance natural-language understanding and response accuracy.
- Integrated Dialogflow with Flask backend APIs to securely access user data and enable context-aware, multi-turn dialogues on banking portal, improving user engagement by 27% through refined training phrases and fallback handling.

PROJECTS

Multi-Agent Financial Report Analyst

05/2026 - Present

- Built multi-agent AI application using **LangGraph** and **RAG** to analyze financial reports, retrieve supporting information, and answer natural-language questions using tool-assisted calculations.
- Orchestrated retrieval, calculation, and verification agents with structured outputs, source citations, and automated correction of unsupported answers.

WhatsApp Fake News Detection

01/2020 - 05/2020

- Developed **BERT-based semantic search** system to verify forwarded messages against trusted news sources, achieving 78.09% accuracy.

Virtual Tourist Guide, Smart India Hackathon

12/2019 - 01/2020

- Created mobile app featuring **CNN classifier** to identify historical landmarks from user-captured images with 92% accuracy.

EDUCATION

Master of Science in Computer Science

08/2021 - 05/2023

University of Southern California, Los Angeles

Bachelor of Technology in Information Technology

09/2017 - 05/2021

Sardar Patel Institute of Technology, Mumbai

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, C

AI/Machine Learning: PyTorch, TensorFlow, Scikit-learn, LangGraph, LangChain, PydanticAI, Hugging Face Transformers, OpenCV, Retrieval-Augmented Generation (RAG), Prompt Engineering

Cloud Technologies: AWS (SageMaker, Bedrock, Lambda, S3, RDS, IAM), Google Cloud Platform (Vertex AI, Cloud Storage, BigQuery, API Gateway), Docker, Kubernetes

Web Development: Node.js, AngularJS, Express.js, React, Django, Flask, Figma

Databases: MySQL, Redis, PostgreSQL, MongoDB

Tools & DevOps: Datadog, Android Studio, gRPC, REST API, GitHub, Apache Kafka

Data Analytics: Power BI, Tableau, Google Data Studio, Spark